

Finite-sample conservatism of the Stringer bound: sharpness at $n = 2$ and conventional confidence levels through $n = 4$

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Abstract

The Stringer bound is widely used to evaluate monetary-unit samples in auditing, but its finite-sample coverage has not been established in general. We prove that, for sample size $n = 2$, the Stringer bound with binomial factors is conservative for every distribution on $[0, 1]$ and every nominal confidence level. The result is sharp: the supremum of the noncoverage probability is α , although no distribution attains it. At $n = 3$ and $n = 4$, exact rational Bernstein certificates prove conservatism at 90%, 95%, and 99% nominal confidence. The proofs establish pointwise domination of a recently validated distribution-free bound for the mean of a bounded variable. At every sample size and every nominal confidence level greater than $1 - e^{-1} \approx 63.2\%$, the Poisson factors used in audit practice dominate the binomial factors coordinatewise and therefore give a pointwise no-smaller Stringer bound. We also give explicit finite-sample counterexamples at larger sample sizes and low nominal confidence levels. At selected nominal confidence levels between 30% and 37%, and for sample sizes between 50 and 400, exact rational certificates show that coverage falls below the nominal level. In searches at 50%, 90%, and 95%, we found no violations; the smallest coverage found agreed with nominal coverage to within 10^{-10} . Finally, we revisit a previous finite-sample certification and identify an index shift in its band-containment argument. The corrected containment event has probability at most $1 - \alpha$, so that argument cannot establish the desired coverage guarantee. These results establish distribution-free finite-sample conservatism at every level for $n = 2$, establish exact conventional-level guarantees through $n = 4$, provide explicit certified failures at larger n , and narrow the unresolved finite-sample regime.

Keywords: Stringer bound; finite-sample coverage; bounded means; monetary-unit sampling; Clopper–Pearson limits.

1 Introduction

Monetary-unit sampling (MUS) draws n dollar units at random from an audited population. The *taint* of a drawn unit is $T \in [0, 1]$, the overstated fraction of the account item associated with that dollar. With taints $t_{(1)} \geq \cdots \geq t_{(n)}$ arranged in nonincreasing order, the Stringer bound [19] at nominal confidence $1 - \alpha$ is

$$\text{SB} = p_n(0) + \sum_{j=1}^n (p_n(j) - p_n(j-1)) t_{(j)}, \quad (1)$$

where $p_n(j)$ is the upper Clopper–Pearson limit for j successes in n trials, i.e. the root of $\sum_{i=0}^j \binom{n}{i} p^i (1-p)^{n-i} = \alpha$, with $p_n(n) = 1$. Audit guidance also tabulates Poisson-based confidence factors for MUS evaluation [1, Appendix C]. Proposition 7 below proves that these factors dominate the binomial factors, at every sample size, whenever nominal confidence exceeds $1 - e^{-1} \approx 63.2\%$. The method is commonly justified by the conjecture that

$$\mathbb{P}(\text{SB} \geq \mu(F)) \geq 1 - \alpha \quad \text{for all } n \text{ and all } F \text{ on } [0, 1], \quad (2)$$

where $T_1, \dots, T_n$ are drawn independently from the common distribution F , with mean $\mu(F)$. This is the standard i.i.d. model used in published analyses of the bound. A National Research Council panel noted that the formulation of the Stringer bound had not been satisfactorily explained [13]. The principal prior results are as follows. Bickel [5] proved (2) for $n = 1$, gave the lower bound $(1 - \alpha)^{n+1}$ for $n \geq 2$ under conditions on F , and derived the expansion $\text{SB} = \bar{T} + c(F)z_{1-\alpha}n^{-1/2} + o_p(n^{-1/2})$ with $c^2(F) \geq \text{Var}(T)$. Pap and van Zuijlen [15] established an almost-sure version of this expansion and concluded that the bound is asymptotically conservative for $\alpha \in (0, \frac{1}{2}]$ and asymptotically *anti-conservative* for $\alpha \in (\frac{1}{2}, 1)$. In [14], they exhibited non-conservatism for uniform-type taints at sub-50% confidence using exact recursions. De Jager, Pap, and van Zuijlen [7] proved that, for taint distributions supported on $\{0, 1\}$, the bound reduces to Clopper–Pearson (where (2) holds exactly) and that Stringer’s coefficients are minimal with that property. A 2021 survey of confidence bounds for bounded means categorized the Stringer bound among methods conjectured to have guaranteed coverage and described its coverage for $\alpha < 0.5$ as unknown [16, p. 8569]. Vlassis and Thomas [18] recently proved the finite-sample validity of Gaffke’s distribution-free mean test [8]. Inversion for observations in $[0, 1]$ gives an upper confidence limit that is a quantile of a uniform Dirichlet average [11, 12]. This development supplies the valid comparison bound used for our $n = 3$ and $n = 4$ results.

This paper makes five contributions.

(1) *A theorem at $n = 2$* (Section 2). The conjecture (2) holds at $n = 2$ for every F and every $\alpha \in (0, 1)$, including $\alpha > 1/2$. Thus the known asymptotic failures do not occur at $n = 2$, and the result is sharp. To our knowledge, this is the first distribution-free finite-sample proof for $n = 2$.

(2) *Conventional confidence levels at $n = 3$ and $n = 4$* (Section 3). At nominal confidence 90%, 95%, and 99%, the binomial Stringer bound pointwise dominates the valid Gaffke upper limit. The comparison is proved by exact rational Bernstein certificates for the relevant uniform-simplex cap volumes. It yields distribution-free finite-sample conservatism at each of these levels for both sample sizes.

(3) *An all- n comparison of the factors used in practice* (Section 4). Whenever nominal confidence exceeds $1 - e^{-1} \approx 63.2\%$, the Poisson factors dominate the binomial factors coordinatewise for every n . A summation-by-parts identity then shows that the Poisson-factor Stringer bound is pointwise no smaller on every observed sample.

(4) *Exact counterexamples at low confidence* (Section 5). Thirty-three certified three-point distributions, each with two nonzero taint values and an atom at zero, give counterexamples at selected nominal confidence levels between 30% and 37%, for sample sizes $n = 50, 100, 200$, and 400. Their coverage is evaluated in exact rational arithmetic with margin certificates. In searches at 50%, 90%, and 95%, no violation was found. At every sample size searched, the smallest coverage found agreed with the nominal level to within 10^{-10} .

(5) *Reassessment of a previous certification* (Section 6). A proposed finite-sample certification [6] contains an index shift in its band-containment argument. We give hand-checkable examples for which the stated containment inequality fails and examine the corrected event.

2 Finite-sample conservatism at $n = 2$

For $n = 2$, the factors are explicit: $p_2(0) = 1 - \sqrt{\alpha}$, $p_2(1) = \sqrt{1 - \alpha}$, $p_2(2) = 1$. Write

$$\beta = \sqrt{\alpha}, \quad \gamma = \sqrt{1 - \alpha} \quad (\beta^2 + \gamma^2 = 1), \quad A = \beta + \gamma - 1, \quad B = 1 - \gamma,$$

so $A, B > 0$ and $A + B = \beta$.

Theorem 1. *Let $\alpha \in (0, 1)$ and let T_1, T_2 be i.i.d. with common distribution F on $[0, 1]$ and mean μ . Then $\mathbb{P}(\text{SB} \geq \mu) \geq 1 - \alpha$. Moreover $\sup_F \mathbb{P}(\text{SB} < \mu) = \alpha$, and the supremum is not attained.*

2.1 Reduction

Substituting $U_i = 1 - T_i$, $w = 1 - \mu = \mathbb{E}[U]$, $m = \min(U_1, U_2)$, $M = \max(U_1, U_2)$ into (1) and telescoping gives

$$\text{SB} = 1 - (Am + BM), \tag{3}$$

so $\text{SB} \geq \mu \iff Am + BM \leq w$. The theorem is implied by the *wedge inequality*: for U_1, U_2 i.i.d. on $[0, 1]$ with mean $w > 0$,

$$\mathbb{P}(Am + BM < w) \geq \gamma^2. \tag{4}$$

(If $w = 0$ then $U \equiv 0$ a.s. and $\text{SB} = 1 \geq \mu$ almost surely.)

If $w > \beta$ the event $\{Am + BM \geq w\}$ is empty, since the left side is at most $A + B = \beta$; if $w = \beta$ it is exactly $\{U_1 = U_2 = 1\}$ (as $A, B > 0$), of probability $\mathbb{P}(U = 1)^2 \leq w^2 = \alpha$ by Markov's inequality.

Henceforth, assume $0 < w < \beta$. The proof has three steps. First, the rectangle lemma converts a hypothetical failure of the wedge inequality into pointwise lower bounds on the survival function G . Second, boundary rectangles relate values of G on opposite sides of u^* . Third, the potential inequality combines those bounds and forces the integral of G to exceed its prescribed value w , producing a contradiction.

2.2 Rectangle bounds

Let $F(x) = \mathbb{P}(U \leq x)$, $G = 1 - F$, and recall $w = \int_0^1 G$. Define

$$u^* = \frac{w}{\beta} \in (0, 1), \quad a_0 = \max\left(0, \frac{w - B}{A}\right), \quad T^\dagger = \min\left(\frac{w}{B}, 1\right),$$

and note $a_0 < u^* < T^\dagger$.

Lemma 2 (Rectangle lemma). *If $0 \leq a \leq b \leq 1$ and $Aa + Bb < w$, then $\mathbb{P}(Am + BM < w) \geq 2F(a)F(b) - F(a)^2$.*

Proof. On $\{U_1 \leq a, U_2 \leq b\} \cup \{U_1 \leq b, U_2 \leq a\}$ we have $m \leq a$, $M \leq b$, hence $Am + BM \leq Aa + Bb < w$ (using $A, B \geq 0$). By inclusion-exclusion (the intersection is $\{U_1 \leq a, U_2 \leq a\}$ since $a \leq b$) the union has probability $2F(a)F(b) - F(a)^2$. $\square$

Assume for contradiction that (4) fails. Then

$$2F(a)F(b) - F(a)^2 < \gamma^2 \quad \text{whenever } 0 \leq a \leq b \leq 1, \quad Aa + Bb < w. \quad (\text{H})$$

Three families of admissible rectangles give pointwise bounds on G :

(i) *Squares.* For $t < u^*$, $(a, b) = (t, t)$ gives $F(t)^2 < \gamma^2$, i.e. $G(t) > 1 - \gamma = B$ for all $t \in [0, u^*)$.

(ii) *Full-height rectangles* (nonvacuous if and only if $w > B$). For $a < a_0$, $(a, 1)$ gives $1 - G(a)^2 < \gamma^2$, i.e. $G(a) > \beta$ on $[0, a_0)$.

(iii) *Boundary pairs.* Let $h(g) = \frac{\beta^2 - g^2}{2(1-g)}$ on $[0, 1)$. For $b \in (u^*, T^\dagger)$ set $\bar{a}(b) = (w - Bb)/A \in (a_0, u^*)$. Every $a' < \bar{a}(b)$ is admissible with b ; letting $a' \uparrow \bar{a}(b)$ and writing $g = G(\bar{a}(b)^-) < 1$, inequality (H) rearranges to

$$G(b) \geq h(G(\bar{a}(b)^-)) \quad \text{for all } b \in (u^*, T^\dagger)$$

because $2(1-g) - \gamma^2 - (1-g)^2 = \beta^2 - g^2$. If $g = 1$ the bound $G(b) \geq 0$ is used instead, consistent with the convention $h_+(1) = 0$ below.

2.3 Potential function and budget

The function $\tilde{k}$ below is chosen to combine the contribution of $G(a)$ with the corresponding lower bound for $G(b(a))$ after the change of variables used below. Let $h_+ = \max(h, 0)$, extended continuously by $h_+(1) = 0$, and define

$$\tilde{k}(g) = g + \frac{A}{B} h_+(g), \quad g \in [B, 1].$$

Lemma 3. $\tilde{k}(g) \geq \beta$ on $[B, 1]$, with equality exactly when $g \in \{B, \beta\}$.

Proof. Since $\beta^2 - B^2 = \alpha - (1 - \gamma)^2 = 2\gamma(1 - \gamma)$, we get $h(B) = B$, hence $\tilde{k}(B) = B + \frac{A}{B}B = A + B = \beta$; also $h(\beta) = 0$ so $\tilde{k}(\beta) = \beta$. Differentiating, $h'(g) = \frac{(1-g)^2 - \gamma^2}{2(1-g)^2}$, so h has its unique critical point at $g = B$ (where $(1-g)^2 = \gamma^2$) and decreases on $(B, 1)$; hence $h \geq 0$ on $[B, \beta]$ (it decreases from $B > 0$ to 0) and $h < 0$ on $(\beta, 1)$. Further $h''(g) = -\gamma^2/(1-g)^3 < 0$; h , and therefore $\tilde{k} = g + \frac{A}{B}h$, is strictly concave on $[B, \beta]$. A strictly concave function with equal endpoint values β satisfies $\tilde{k} \geq \beta$ on $[B, \beta]$, with equality only at the endpoints. On $(\beta, 1]$, $h_+ = 0$ and $\tilde{k}(g) = g > \beta$. $\square$

Proof of Theorem 1. Assume (H). The substitution $t = b(a) = (w - Aa)/B$ is a decreasing bijection from $[a_0, u^*]$ onto $[u^*, T^\dagger]$, with $|dt| = (A/B) da$ (endpoints: $b(u^*) = u^*$; $b(a_0) = 1 = T^\dagger$ if $w \geq B$, $b(0) = w/B = T^\dagger$ if $w \leq B$). Using $w = \int_0^1 G \geq \int_0^{a_0} G + \int_{a_0}^{u^*} G + \int_{u^*}^{T^\dagger} G$, we bound the tail with (iii) (the left-limit $G(a^-)$ agrees with $G(a)$ for all but countably many a , so the integrals coincide):

$$\int_{u^*}^{T^\dagger} G = \frac{A}{B} \int_{a_0}^{u^*} G(b(a)) da \geq \frac{A}{B} \int_{a_0}^{u^*} h_+(G(a)) da.$$

By (i), $G > B$ pointwise on $[a_0, u^*)$, so Lemma 3 applies under the integral:

$$w \geq \int_0^{a_0} G + \int_{a_0}^{u^*} \tilde{k}(G(a)) da \geq \beta a_0 + \beta(u^* - a_0) = \beta u^* = w,$$

where $\int_0^{a_0} G \geq \beta a_0$ by (ii). Equality is forced throughout; we exhibit a strict inequality in every case:

- If $a_0 > 0$: (ii) is pointwise strict on $[0, a_0)$, so $\int_0^{a_0} G > \beta a_0$.
- If $a_0 = 0$ and $G \neq \beta$ on a positive-measure subset of $(0, u^*)$: since $G > B$ there, $\tilde{k}(G) > \beta$ strictly (Lemma 3), so $\int_0^{u^*} \tilde{k}(G) > \beta u^*$.
- If $a_0 = 0$ and $G = \beta$ a.e. on $(0, u^*)$: for each $b \in (u^*, T^\dagger)$ choose $a' < \bar{a}(b)$ with $G(a') = \beta$ (possible, as such a' have full measure in a nonempty interval); the *strict* inequality (H) at the admissible corner (a', b) reads $(1 - \beta)(1 + \beta - 2G(b)) < \gamma^2 = (1 - \beta)(1 + \beta)$, forcing $G(b) > 0$ for every $b \in (u^*, T^\dagger)$; hence $\int_{u^*}^{T^\dagger} G > 0$ and $w \geq \int_0^{u^*} G + \int_{u^*}^{T^\dagger} G > \beta u^* = w$.

Each case yields $w > w$, a contradiction. Thus (H) is impossible, proving (4) and the coverage statement.

Sharpness. Distributions supported on $\{0, u\}$, with $F = (1 - q)\delta_0 + q\delta_u$, give failure probability q^2 when $q < \beta$ (the pair (u, u) , since $\beta u > qu = w$) and additionally the mixed pairs when $q < B$ (since $Bu > qu$), totalling $1 - (1 - q)^2$; letting $q \uparrow \beta$ and $q \uparrow B$ respectively yields failure probabilities approaching $\beta^2 = \alpha$ and $1 - \gamma^2 = \alpha$. Hence $\sup_F \mathbb{P}(\text{SB} < \mu) = \alpha$.

Non-attainment. If some distribution F attained the supremum, then $\mathbb{P}(Am + BM > w) = \alpha$. Such a distribution must have $0 < w < \beta$, since the strict failure event is empty in the boundary regimes handled above. The coverage result just proved gives $\mathbb{P}(Am + BM < w) \geq \gamma^2$; since $\alpha + \gamma^2 = 1$, attainment would therefore require

$$\mathbb{P}(Am + BM < w) = \gamma^2, \quad \mathbb{P}(Am + BM = w) = 0. \quad (5)$$

The rectangle bounds consequently hold with weak inequalities. Repeating the budget argument in that form forces equality at every step. Equality on the first interval gives $G = \beta$ almost everywhere on $(0, a_0)$, and equality in the potential bound permits $G \in \{B, \beta\}$ almost everywhere on (a_0, u^*) . Because G is nonincreasing, there is a $c \in [a_0, u^*]$ such that its almost-everywhere level is β before c and B after c . Set $d = b(c) = (w - Ac)/B$. Equality in the tail comparison, together with $h_+(\beta) = 0$ and $h_+(B) = B$, gives $G(b(a)) = 0$ for almost every $a \in (a_0, c)$ and $G(b(a)) = B$ for almost every $a \in (c, u^*)$. Because b is decreasing, this is the corresponding switch at d on the other side of u^* . Equality in the discarded tail and right-continuity then force

$$G(x) = \begin{cases} \beta, & 0 \leq x < c, \\ B, & c \leq x < d, \\ 0, & d \leq x \leq 1, \end{cases} \quad Ac + Bd = w. \quad (6)$$

The change of variables gives $c \leq u^* \leq d$. Every distribution in (6) places positive probability on the boundary $Am + BM = w$. If $0 < c < d$, it has masses A and B at c and d , so the mixed pair (c, d) has probability $2AB > 0$. If $c = 0 < d$, the atoms at 0 and d have masses

$1 - B$ and B , so their mixed pair has positive probability. The remaining case $c = d$ forces $c = d = u^*$, and the pair (u^*, u^*) has probability $\beta^2 > 0$. Each case contradicts (5). Hence the supremum is not attained. $\square$

Remark 4 (Relation to known inequalities). The equal-weights case of (4), which bounds $\mathbb{P}(X_1 + X_2 \geq t)$ for nonnegative i.i.d. variables with a given mean, is the classical sharp inequality of Hoeffding and Shrikhande [9] (extended to general k in [10]); the wedge inequality is its order-statistic-weighted analogue, with genuinely unequal weights except at $\alpha = 16/25$, where $A = B$. The sharp-bound literature for order statistics and L-statistics under moment conditions (see [4]) concerns expectations or distribution functions of single order statistics, not mean-constrained tail probabilities of weighted combinations; we are not aware of a prior result implying (4). Other distribution-free upper bounds for bounded means include those of Anderson [2] and Gaffke [8]; the latter’s finite-sample validity for independent variables was proved recently [18]. Those results are not used here, and the proof of Theorem 1 is self-contained.

3 Conventional confidence levels at $n = 3$ and $n = 4$

The recent validity proof for Gaffke’s mean test [18] provides a second route to finite-sample coverage. For a real random variable X , write $Q_q(X) = \inf\{z : \Pr(X \leq z) \geq q\}$. Given n observed taints $t_1, \dots, t_n$, define

$$G_\alpha(t_1, \dots, t_n) = Q_{1-\alpha} \left(D_0 + \sum_{i=1}^n t_i D_i \right), \quad (D_0, \dots, D_n) \sim \text{Dirichlet}(1, \dots, 1).$$

This is the one-sided Gaffke–Learned–Miller–Thomas upper limit [8, 11, 12]; it has coverage at least $1 - \alpha$ for independent observations from every distribution on $[0, 1]$. We prove that Stringer is pointwise no smaller at three conventional confidence levels for $n = 3$ and $n = 4$.

3.1 The case $n = 3$

Theorem 5. *Let $n = 3$ and let T_1, T_2, T_3 be i.i.d. with common distribution F on $[0, 1]$ and mean μ . If $\alpha \in \{0.10, 0.05, 0.01\}$, then*

$$\text{SB} \geq G_\alpha(T_1, T_2, T_3) \quad \text{for every observed sample,} \quad \Pr_F(\text{SB} \geq \mu) \geq 1 - \alpha.$$

Thus the binomial Stringer bound is distribution-free conservative at 90%, 95%, and 99% nominal confidence when $n = 3$.

Proof. Put

$$a = 1 - p_3(2), \quad b = p_3(2) - p_3(1), \quad c = p_3(1) - p_3(0), \quad d = p_3(0).$$

For ascending sample taints $t_1 \leq t_2 \leq t_3$, $\text{SB} = at_1 + bt_2 + ct_3 + d$. The four coefficients are positive and sum to one. Unless $t_1 = 1$, affine equivariance permits subtraction of t_1 and division by $1 - t_1$. It therefore suffices to use the four knots $(0, x, y, 1)$, where $0 \leq x \leq y \leq 1$, and the Stringer threshold

$$s = bx + cy + d.$$

By exchangeability of the Dirichlet weights, pointwise domination of G_α is equivalent to

$$V(x, y) := \Pr_D\{xD_1 + yD_2 + D_3 > s\} \leq \alpha. \quad (7)$$

The uniform-simplex cap formula gives

$$V(x, y) = \begin{cases} \frac{(1-s)^3}{(1-x)(1-y)}, & s \geq y, \\ \frac{1}{1-y} \left\{ \frac{(1-s)^3}{1-x} - \frac{(y-s)^3}{y(y-x)} \right\}, & x \leq s \leq y, \\ 1 - \frac{s^3}{xy}, & s \leq x, \end{cases} \quad (8)$$

with boundary values defined by continuity. The Clopper–Pearson equations imply

$$(a+b+c)^3 = \alpha, \quad (b+c+d)^3 = 1-\alpha, \quad (a+b)^2\{3-2(a+b)\} = \alpha. \quad (9)$$

We verify (7) in the three regions of (8). If $s \leq x$, put $g = b+c+d = (1-\alpha)^{1/3}$. The exact factor enclosures certify $b, c \leq g/3$ at each of the three values of α . Weighted AM–GM therefore gives

$$\frac{s}{g} = \frac{b}{g}x + \frac{c}{g}y + \frac{d}{g} \geq x^{b/g}y^{c/g} \geq (xy)^{1/3},$$

and the third line of (8) is at most α .

For $s \geq y$, set $X = 1-x$, $Y = 1-y$, $L = 1-s = a+bX+cY$, and $A = 1-b-c = a+d$. The domain is the triangle with vertices

$$\left(\frac{a}{A}, \frac{a}{A}\right), \quad (1, 1), \quad \left(1, \frac{a+b}{1-c}\right).$$

On that triangle the desired inequality is

$$P_A(X, Y) := \alpha XY - (a+bX+cY)^3 \geq 0. \quad (10)$$

After an affine map from the standard triangle, all degree-three Bernstein coefficients of P_A are positive except the coefficient at $(X, Y) = (1, 1)$, which is zero by the first identity in (9). The coefficient signs are certified by exact rational intervals as described below.

It remains to consider $x \leq s \leq y$. Write $\rho = s-x \geq 0$ and $\tau = y-s \geq 0$. Solving for x, y shows that the (ρ, τ) domain is the quadrilateral with vertices

$$(0, 0), \quad \left(\frac{d}{1-c}, 0\right), \quad (c+d, a+b), \quad \left(0, \frac{a}{1-b}\right).$$

Triangulate it along the diagonal from $(0, 0)$ to $(c+d, a+b)$. Differentiating the middle line of (8) and cancelling common factors gives

$$\frac{\partial V}{\partial y} = \frac{Q(x, y)}{y^2(1-x)(y-x)^2},$$

where Q is a degree-five polynomial. On both triangles, every degree-five Bernstein coefficient of A^3Q is nonnegative. Three on each triangle are structural zeros, and all remaining coefficients are strictly positive. Hence V is nondecreasing in y at fixed x . Increasing y preserves both middle-region inequalities, so it is enough to take the limit $y \uparrow 1$.

At $y = 1$, put $q = 1 - x$. The constraint is $q \geq q_0 := a/(1 - b)$, and direct simplification yields

$$V(x, 1) = \frac{(a + bq)^2 \{3q - (a + bq)(1 + q)\}}{q^2}.$$

After mapping $[q_0, 1]$ to $[0, 1]$, the degree-four Bernstein coefficients of

$$\alpha q^2 - (a + bq)^2 \{3q - (a + bq)(1 + q)\}$$

are positive except the coefficient at $q = 1$, which is zero by the final identity in (9). This completes (7).

For completeness, Table 1 reports the smallest positive Bernstein lower bound in each exact check. The factors are enclosed on a dyadic grid of width 2^{-120} ; endpoint signs of their binomial CDF equations and every subsequent coefficient sign are evaluated with integer or rational arithmetic. The symbolic derivation starts from (8) and regenerates all coefficient formulas. Both the derivation and sign certificate are reproduced by the repository commands `make n3-formula-check` and `make n3-certificate-check`. Thus this is an exact computer-assisted proof, not a grid search. Finally, validity of G_α gives the stated coverage inequality. $\square$

α	region $s \geq y$	middle triangle 1	middle triangle 2	$y = 1$ boundary
0.01	1.04×10^{-7}	9.45×10^{-5}	1.37×10^{-9}	3.67×10^{-8}
0.05	1.63×10^{-5}	1.38×10^{-4}	7.52×10^{-8}	4.40×10^{-6}
0.10	1.45×10^{-4}	1.04×10^{-4}	3.27×10^{-7}	3.28×10^{-5}

Table 1: Smallest positive exact rational lower bound among the relevant Bernstein coefficients in the $n = 3$ certificate. Displayed values are decimal approximations; the complete rational bounds are in the machine-readable certificate.

3.2 The case $n = 4$

The same comparison can be carried out one dimension higher. The additional dimension changes the cap geometry substantially: after clearing boundary factors, two degree-six polynomials must be certified on six tetrahedra.

Theorem 6. *Let $n = 4$ and let $T_1, \dots, T_4$ be i.i.d. with common distribution F on $[0, 1]$ and mean μ . If $\alpha \in \{0.10, 0.05, 0.01\}$, then*

$$\text{SB} \geq G_\alpha(T_1, \dots, T_4) \quad \text{for every observed sample,} \quad \Pr_F(\text{SB} \geq \mu) \geq 1 - \alpha.$$

Thus the binomial Stringer bound is distribution-free conservative at 90%, 95%, and 99% nominal confidence when $n = 4$.

Proof. Write $p_j = p_4(j)$ and put

$$a = 1 - p_3, \quad b = p_3 - p_2, \quad c = p_2 - p_1, \quad d = p_1 - p_0, \quad e = p_0.$$

For ascending taints $t_1 \leq t_2 \leq t_3 \leq t_4$, $SB = at_1 + bt_2 + ct_3 + dt_4 + e$. The five coefficients are positive and sum to one. As in the $n = 3$ proof, affine equivariance reduces the sample knots to

$$(0, x, y, z, 1), \quad 0 \leq x \leq y \leq z \leq 1, \quad s = bx + cy + dz + e.$$

It is therefore enough to prove that

$$V(x, y, z) := \Pr_D\{xD_1 + yD_2 + zD_3 + D_4 > s\} \leq \alpha, \quad (11)$$

where $(D_0, \dots, D_4) \sim \text{Dirichlet}(1, \dots, 1)$. Put

$$H_1 = \frac{(1-s)^4}{(1-x)(1-y)(1-z)}, \quad H_z = \frac{(z-s)^4}{z(z-x)(z-y)(1-z)}, \quad H_y = \frac{(y-s)^4}{y(y-x)(z-y)(1-y)}.$$

The uniform-simplex cap formula is

$$V = \begin{cases} H_1, & s \geq z, \\ H_1 - H_z, & y \leq s \leq z, \\ H_1 - H_z + H_y, & x \leq s \leq y, \\ 1 - \frac{s^4}{xyz}, & s \leq x, \end{cases} \quad (12)$$

with boundary values defined by continuity. The factor equations give

$$\begin{aligned} \alpha &= (1-e)^4, \\ \alpha &= (1-d-e)^3\{1+3(d+e)\}, \\ \alpha &= (1-c-d-e)^2\{1+2(c+d+e)+3(c+d+e)^2\}, \\ 1-\alpha &= (b+c+d+e)^4. \end{aligned} \quad (13)$$

First suppose $s \leq x$ and set $g = b + c + d + e = p_3$. The exact factor enclosures certify $b, c, d \leq g/4$ at each stated value of α . Weighted AM-GM gives

$$\frac{s}{g} = \frac{b}{g}x + \frac{c}{g}y + \frac{d}{g}z + \frac{e}{g} \geq x^{b/g}y^{c/g}z^{d/g} \geq (xyz)^{1/4}.$$

The last identity in (13) now makes the last line of (12) at most α .

For the other three regions, clear the positive denominators in (12). In the region $s \geq z$, the required residual is

$$P_A = \alpha(1-x)(1-y)(1-z) - (1-s)^4.$$

In the region $y \leq s \leq z$, use

$$\begin{aligned} P_B &= \alpha(1-x)(1-y)(1-z)z(z-x)(z-y) \\ &\quad - (1-s)^4z(z-x)(z-y) + (z-s)^4(1-x)(1-y). \end{aligned}$$

Exact polynomial division, using the second identity in (13), gives $P_B = (1 - z)R_B$, where R_B has degree six. In the region $x \leq s \leq y$, the corresponding cleared residual is

$$\begin{aligned} P_C = & \alpha(1 - x)(1 - y)(1 - z)z(z - x)(z - y)y(y - x) \\ & - (1 - s)^4 z(z - x)(z - y)y(y - x) \\ & + (z - s)^4 (1 - x)(1 - y)y(y - x) \\ & - (y - s)^4 (1 - x)(1 - z)z(z - x). \end{aligned}$$

The third identity in (13) gives $P_C = (1 - z)(1 - y)(z - y)R_C$, with R_C of degree six. Every extracted factor is nonnegative on the ordered-knot domain.

It remains to certify P_A, R_B, R_C . Define

$$\begin{aligned} r_0 = (0, 0, 0), \quad r_1 = \left(0, 0, \frac{e}{1 - d}\right), \quad r_2 = \left(0, \frac{e}{1 - c - d}, \frac{e}{1 - c - d}\right), \\ r_3 = (t, t, t), \quad t = \frac{e}{1 - b - c - d}, \quad v_{001} = (0, 0, 1), \\ q_2 = \left(0, \frac{d + e}{1 - c}, 1\right), \quad q_3 = (u, u, 1), \quad u = \frac{d + e}{1 - b - c}, \\ v_{011} = (0, 1, 1), \quad q_4 = \left(\frac{c + d + e}{1 - b}, 1, 1\right). \end{aligned}$$

Intersecting $0 \leq x \leq y \leq z \leq 1$ successively with the planes $s = z$, $s = y$, and $s = x$ gives the following tetrahedralizations:

$$\begin{aligned} s \geq z : & \quad [r_0, r_1, r_2, r_3], \\ y \leq s \leq z : & \quad [r_3, r_2, r_1, v_{001}], \quad [r_3, q_2, q_3, v_{001}], \quad [r_3, q_2, r_2, v_{001}], \\ x \leq s \leq y : & \quad [r_3, q_2, r_2, v_{011}], \quad [q_3, r_3, q_4, v_{011}], \quad [q_3, r_3, q_2, v_{011}]. \end{aligned}$$

The listed tetrahedra meet only on common faces and cover the respective convex polytopes; this also follows directly by slicing at the common diagonals $[r_3, v_{001}]$ and $[r_3, v_{011}]$.

Map each tetrahedron affinely to the standard tetrahedron. The exact symbolic calculation finds one structural-zero coefficient in the degree-four Bernstein expansion of P_A . For R_B , the three degree-six expansions have respectively 10, 5, and 10 structural zeros; for R_C , they have 5, 10, and 10. These coefficients vanish identically over $\mathbb{Q}(b, c, d, e)$. Every remaining coefficient has a strictly positive exact rational interval enclosure at each stated level. Table 2 reports the smallest positive lower bounds by region. Factor endpoints lie on a dyadic grid of width 2^{-120} , their binomial-CDF signs are checked with integer arithmetic, and all polynomial, vertex, and Bernstein calculations thereafter use exact rational arithmetic. The symbolic and numerical layers are regenerated by **make n4-structure-check** and **make n4-certificate-check**. Hence $P_A, R_B, R_C \geq 0$, and (11) follows in all four regions. Validity of G_α then proves the coverage statement. $\square$

α	region $s \geq z$	region $y \leq s \leq z$	region $x \leq s \leq y$
0.01	3.10×10^{-10}	4.78×10^{-14}	5.84×10^{-12}
0.05	3.47×10^{-7}	2.66×10^{-10}	4.60×10^{-9}
0.10	7.74×10^{-6}	1.18×10^{-8}	7.93×10^{-8}

Table 2: Smallest positive exact rational lower bound among the relevant Bernstein coefficients in the $n = 4$ certificate. Displayed values are decimal approximations; the exact bounds and all structural-zero indices are machine-readable.

4 Poisson factors at conventional audit confidence levels

For $0 \leq j \leq n$, let $p_n^B(j)$ denote the binomial factor used above, and let

$$p_n^P(j) = \frac{\lambda_j}{n}, \quad \sum_{i=0}^j e^{-\lambda_j} \frac{\lambda_j^i}{i!} = \alpha,$$

denote its Poisson analogue. The Poisson factors are not truncated at one; as in audit factor tables, a factor may exceed one.

Proposition 7 (All- n factor domination). *If $0 < \alpha < e^{-1}$, then*

$$p_n^P(j) \geq p_n^B(j), \quad j = 0, \dots, n, \quad n \geq 1.$$

Consequently, the Poisson-factor Stringer bound is at least the binomial-factor Stringer bound for every observed sample.

Proof. Write $B(j; n, p) = \Pr\{\text{Bin}(n, p) \leq j\}$ and $P(j; \lambda) = \Pr\{\text{Pois}(\lambda) \leq j\}$. Anderson and Samuels [3, Corollary 2.1 and Section 4.2] show that

$$B(j; n, \lambda/n) < P(j; \lambda) \quad \text{when } j < \lambda - 1 \text{ and } \lambda < n.$$

Because $P(j; \lambda_j) = \alpha < e^{-1} = P(0; 1)$, their Theorem 4.1 gives $\lambda_j > j + 1$. If $\lambda_j < n$, the displayed inequality yields $B(j; n, p_n^P(j)) < \alpha$; since $B(j; n, p)$ is decreasing in p , its root satisfies $p_n^B(j) < p_n^P(j)$. If $\lambda_j \geq n$, the conclusion follows instead from $p_n^P(j) \geq 1 \geq p_n^B(j)$.

For the pointwise comparison, put $d_j = p_n^P(j) - p_n^B(j) \geq 0$. If $t_{(1)} \geq \dots \geq t_{(n)}$, summation by parts gives

$$\text{SB}_P - \text{SB}_B = d_0(1 - t_{(1)}) + \sum_{j=1}^{n-1} d_j(t_{(j)} - t_{(j+1)}) + d_n t_{(n)} \geq 0.$$

□

Corollary 8. *The Poisson-factor Stringer bound is distribution-free conservative at $n = 2$ whenever $0 < \alpha < e^{-1}$, and at $n = 3$ and $n = 4$ for $\alpha \in \{0.10, 0.05, 0.01\}$.*

Proof. Combine Proposition 7 with Theorems 1, 5, and 6. □

Thus the factor comparison previously checked numerically at 90% and 95% is analytic and holds at every sample size. In particular, Theorem 1 implies finite-sample conservatism of the Poisson-factor Stringer bound at $n = 2$ throughout this confidence range. Theorems 5 and 6 give the same conclusion at $n = 3$ and $n = 4$ for the three certified conventional levels. More generally, any distributional class for which the binomial-factor bound is proved conservative inherits the same conclusion for the Poisson-factor bound. Proposition 7 does not, by itself, settle coverage in the remaining cases, because the binomial-factor conjecture remains unresolved there.

Proposition 9 (One nonzero taint value). *If F is supported on $\{0, v\}$ with $v \in (0, 1]$, then, for every n and α , $\mathbb{P}(\text{SB} \geq \mu) \geq 1 - \alpha$.*

Proof. Let K denote the number of nonzero taints. Then $K \sim \text{Bin}(n, q)$,

$$\text{SB} = (1 - v)p_n(0) + v p_n(K) \geq v p_n(K), \quad \mu = qv.$$

The Clopper–Pearson coverage event $\{p_n(K) \geq q\}$ has probability at least $1 - \alpha$; on that event, $\text{SB} \geq vq = \mu$. $\square$

5 Exact counterexamples at low confidence

Theorem 1 covers every confidence level, so the asymptotic anti-conservatism at $\alpha > 1/2$ [15] must manifest itself only at larger n . We give explicit finite-sample counterexamples. Each counterexample below is certified: its distributional parameters are exact dyadic rationals, its coverage is computed as an exact rational by full multinomial enumeration, and every comparison $\text{SB} \geq \mu$ is made with rational factor intervals. Their endpoints lie on a dyadic grid with denominator 2^{80} , and the sign of the binomial CDF at each endpoint is evaluated exactly with integer arithmetic. The table rows are regenerated from the version-2 certificate files by `summarize_certificates.py`.

confidence	n	smallest certified coverage	certified examples
30%	50	0.29815	3
32%	100	0.30956	9
35%	200	0.33729	1
35%	400	0.32630	1
37%	200	0.36079	6
37%	400	0.36777	13

Table 3: Certified finite-sample violations (binomial factors, three-point taint distributions with two nonzero taint values). The listed results illustrate the onset of failure in the examined parameter ranges.

A representative certificate has nominal confidence 30% ($\alpha = 0.7$) and $n = 50$. Set

$$\begin{aligned} v_1 &= 1, & q_1 &= \frac{8984394653678481}{2^{55}}, \\ v_2 &= \frac{8384891858230937}{2^{54}}, & q_2 &= \frac{5642023711000999}{2^{53}}, \\ & & q_0 &= \frac{4476307521281491}{2^{55}}. \end{aligned}$$

The complete distribution is $\mathbb{P}(T = v_1) = q_1$, $\mathbb{P}(T = v_2) = q_2$, and $\mathbb{P}(T = 0) = q_0$. Thus the two nonzero taint values are 1 and approximately 0.465455, and the corresponding probabilities are approximately 0.249367 and 0.626390. These parameters appear in the third record of the $n = 50$, $\alpha = 0.7$ certificate (`certified-alpha0.7-n50.json`). The certificate prints the exact rational coverage in full. Its decimal value is approximately 0.2988790828, below the nominal level 0.30. The minimum comparison gap certified by rational interval arithmetic is approximately 2.10×10^{-14} ; each factor interval has width at most $2^{-80} \approx 8.27 \times 10^{-25}$. Pap and van Zuijlen [14] established non-conservatism for uniform-type taints through exact recursions; the finite-support examples here are readily reproducible and illustrate the onset of failure in the examined parameter ranges.

By contrast, searches at nominal confidence levels 50%, 90%, and 95% found no violation. Grid-plus-optimizer searches examined distributions with two nonzero taint values ($n \leq 100$ at 95% and 90%; $n \leq 300$ at 50%) and distributions with three nonzero taint values ($n \leq 25$ at 95%), in each case allowing mass at zero. At every sample size searched, the smallest coverage found agreed with the nominal level to within 10^{-10} . Near-minimizers occurred as the largest taint approached 1, consistent with Proposition 9, the $\{0, 1\}$ -support minimality result of [7], and the conjectured coverage infimum $1 - \alpha$. For the Poisson factors, the smallest coverage found in the corresponding searches remained above nominal. At 95%, for example, the values were approximately 0.9716 ($n = 10$), 0.9573 ($n = 50$), and 0.9549 ($n = 100$), a positive numerical cushion that decreases over the listed sample sizes.

6 Reassessment of a previous finite-sample certification

Let

$$C_n(F) = \mathbb{P}_F(\text{SB} \geq \mu(F))$$

denote the coverage probability. In this section, $t_{1:n} \leq \dots \leq t_{n:n}$ are the sample taints arranged in nondecreasing order, with $t_{0:n} = 0$ and $t_{n+1:n} = 1$. Let $q_n(i)$ be the lower one-sided Clopper–Pearson limit for i successes, with $q_n(0) = 0$. The lower and upper limits used here satisfy $q_n(i) = 1 - p_n(n - i)$.

Bimpeh [6, pp. 76–79] writes the Stringer bound as $\text{SB} = \int_0^1 [1 - \hat{F}_{n,L}(t)] dt$, where the stepwise band is defined by

$$\hat{F}_{n,L}(t) = q_n(i - 1) \quad \text{for } t_{i-1:n} \leq t < t_{i:n}, \quad 1 \leq i \leq n + 1.$$

Let $V_{1:n} \leq \dots \leq V_{n:n}$ denote the order statistics of n independent $\text{Unif}(0, 1)$ random variables. Using the probability integral transform and the reversal symmetry of uniform order statistics,

Bimpeh proposes the lower bound

$$C_n(F) \geq \bar{P}_n := \mathbb{P}(V_{i:n} \leq p_n(i), 1 \leq i \leq n). \quad (14)$$

Bimpeh evaluates $\bar{P}_n$ using Bolshev's recursion for uniform order-statistic boundary-crossing probabilities [17, pp. 366–367]. Because the computed values satisfy $\bar{P}_n \geq 1 - \alpha$ for $n \leq 11$ when $\alpha = 0.05$, the thesis concludes that (2) holds in that range. It also notes that

$$\bar{P}_2 = 2\sqrt{1 - \alpha} - (1 - \alpha) \geq 1 - \alpha$$

for every α and claims a proof for $n \leq 2$.

Proposition 10. *Inequality (14) fails for both continuous and atomic distributions F .*

Proof. Let $\alpha = 0.05$ and let F be the mixture assigning probability 0.99 to a uniform distribution on $(0.99, 1)$ and probability 0.01 to a uniform distribution on $(0, 0.01)$, so $\mu = 0.9851$. For $n = 1$, $\text{SB} = 0.95 + 0.05 T_1$ and $\bar{P}_1 = 1$. Moreover, $\text{SB} < \mu$ if and only if $T_1 < 0.702$, an event of probability 0.01. Thus $C_1(F) = 0.99 < 1 = \bar{P}_1$. For $n = 2$, $\bar{P}_2 = 2\sqrt{0.95} - 0.95 \approx 0.99936$. A sample containing one taint from each mixture component fails, as does a sample containing two taints from the lower component. Therefore $C_2(F) = 1 - 2(0.99)(0.01) - 0.01^2 = 0.9801 < \bar{P}_2$.

For an atomic example at $n = 5$, let $\mathbb{P}(T = 1) = \frac{1}{2}$ and $\mathbb{P}(T = \frac{1}{10}) = \mathbb{P}(T = 0) = \frac{1}{4}$. Its exact coverage is $C_5(F) = \frac{31}{32} = 0.96875 < \bar{P}_5 \approx 0.98746$. The failure event is precisely the event of drawing no taint-1 unit, which has probability 2^{-5} . $\square$

The issue is an index shift. Pointwise domination $F \geq \hat{F}_{n,L}$ on $[0, 1]$ requires $F(t_{i:n}) \geq q_n(i)$ for all i , because $q_n(i)$ is the band's value at the jump point $t_{i:n}$. The cited argument instead uses the left limit $q_n(i - 1)$. In particular, it omits the constraint $F(t_{n:n}) \geq q_n(n) = \alpha^{1/n}$. Correcting the index does not recover the claimed coverage guarantee.

Proposition 11. *For every continuous F and every n , the corrected containment probability satisfies $\mathbb{P}(F(t_{i:n}) \geq q_n(i) \ \forall i) = \mathbb{P}(V_{i:n} \geq q_n(i) \ \forall i) \leq 1 - \alpha$.*

Proof. For continuous F , the probability integral transform gives $F(t_{i:n}) = V_{i:n}$ in distribution. The corrected event includes its $i = n$ constraint, and $q_n(n) = \alpha^{1/n}$. Hence

$$\mathbb{P}(V_{i:n} \geq q_n(i) \ \forall i) \leq \mathbb{P}(V_{n:n} \geq \alpha^{1/n}) = 1 - \alpha,$$

because $\mathbb{P}(V_{n:n} < x) = x^n$. $\square$

Thus the corrected pointwise-containment event has probability no greater than nominal coverage. The integral inequality $\int \hat{F}_{n,L} \leq \int F$ may hold on a larger event than pointwise domination, but the cited containment argument does not establish that fact. Our implementation reproduces the values of $\bar{P}_n$ and Bimpeh's Table 5.1 exactly; those values are boundary-crossing probabilities, not established coverage bounds. Consequently, the cited argument does not establish finite-sample conservatism at $n = 2$. Theorem 1 supplies that result directly.

7 Discussion

Three observations suggest directions for future work. First, the reformulation behind (3) generalizes: with $u_{(1)} \leq \dots \leq u_{(n)}$ the increasing order statistics of $U_i = 1 - T_i$,

$$\text{SB} = 1 - \sum_{j=1}^n e_j u_{(j)}, \quad e_j = p_n(j) - p_n(j-1),$$

so (2) is a family of weighted order-statistic exceedance inequalities, with the audit-specific confidence-factor structure encoded in the positive increments e_j . Second, the $n = 2$ proof combines rectangle lower bounds on the covering event, a survival-integral budget, and a concave potential with equal endpoint values. The $n = 3$ and $n = 4$ proofs use a different route: they compare Stringer with a valid Dirichlet-average bound and certify the resulting simplex-cap inequalities. In higher dimensions, the analogous pointwise comparison is a concrete geometric problem for caps of a uniform simplex. At 90%, 95%, and 99% confidence, $n = 5$ is now the first unresolved sample size. Third, the near-equality observed at every searched n , together with the certified failures at several nominal confidence levels below 50%, suggests that (2) may hold whenever nominal confidence is at least 50%, equivalently whenever $\alpha \leq 1/2$, with coverage infimum $1 - \alpha$ at every n . This is a concrete conjecture for future work.

From an auditing perspective, the binomial Stringer bound has no uniform safety margin above nominal coverage: at $n = 2$, its coverage infimum is $1 - \alpha$. Theorems 5 and 6 supply exact guarantees at the three conventional levels for $n = 3$ and $n = 4$. At conventional confidence levels, the Poisson version pointwise dominates the binomial version by Proposition 7; the reported Poisson searches also retain a positive but decreasing numerical cushion. The certified examples show that anti-conservatism can occur at several nominal confidence levels below 50%, at sample sizes $n = 50, 100, 200$, and 400. Coverage for ordinary audit sample sizes at conventional confidence remains an open problem, so the search evidence alone does not justify a change in audit guidance.

Reproducibility

The repository contains exact rational coverage calculations, margin certificates, exact-sign dyadic enclosures for the binomial confidence factors, exact rational Bernstein certificates for Theorems 5 and 6, symbolic regeneration checks for their simplex-cap formulas, high-precision Poisson calculations, search logs, symbolic checks of the identities used in the $n = 2$ proof, and stress tests over approximately 200,000 distributions. It also contains records of two separate adversarial review passes.

The canonical source repository is <https://github.com/RichieSater/stringer-bound>. Release v1.0.0 is preserved on Zenodo at <https://doi.org/10.5281/zenodo.21850820>; the version-independent DOI, which resolves to the latest archived release, is <https://doi.org/10.5281/zenodo.21850819>.

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